



Catalytic pyrolysis of plastic wastes with two different types of catalysts: ZSM-5 zeolite and Red Mud

A. López^{a,*}, I. de Marco^a, B.M. Caballero^a, M.F. Laresgoiti^a, A. Adrados^a, A. Aranzabal^b

^a Chemical and Environmental Engineering Department, School of Engineering of Bilbao, University of the Basque Country, Alda, Urquijo s/n, 48013-Bilbao, Spain

^b Chemical Engineering Department, Faculty of Sciences and Technology, University of the Basque Country, P.O. Box 644, 48080-Leioa, Spain

ARTICLE INFO

Article history:

Received 11 January 2011

Received in revised form 16 March 2011

Accepted 24 March 2011

Available online 1 April 2011

Keywords:

Feedstock recycling

Catalytic pyrolysis

Plastic wastes

ZSM-5 zeolite

Red Mud

ABSTRACT

The influence of ZSM-5 zeolite and Red Mud in the pyrolysis of plastic wastes has been studied. Both catalysts have been thoroughly characterized; the zeolite shows weak and strong acid sites and great BET surface area ($412.0 \text{ m}^2 \text{ g}^{-1}$), while Red Mud contains lower acidity, with also weak and strong acid sites, meso-macropores and BET surface area of $27.49 \text{ m}^2 \text{ g}^{-1}$. Both catalysts have been tested in pyrolysis of a mixture of plastics which resembles municipal plastic wastes, at 440 and 500 °C in a 3.5 dm^3 semi-batch reactor. The results have been compared with those of the thermal process. It has been proved that ZSM-5 zeolite has a strong effect in the characteristics and distribution of pyrolysis products. It generates at both temperatures a greater proportion of gases and liquids with a higher content of aromatics than without catalyst. Red Mud needs higher temperatures than ZSM-5 zeolite to exert a catalytic effect in pyrolysis, since similar results to those obtained without catalyst are obtained at 440 °C, while at 500 °C a higher yield of gases and a greater proportion of aromatics in the liquids is obtained.

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1. Introduction

Research on feedstock recycling of plastic wastes is receiving increasing attention in the last years. The unstoppable generation of plastic wastes, the market limitations to some mechanically recycled plastic goods, and the possibility of obtaining valuable products from complex plastic wastes that cannot be recycled by simple mechanical recycling, are some of the driving forces which promote chemical recycling research.

Feedstock recycling includes several techniques such as depolymerisation, thermal and thermo-catalytic decomposition and the utilization of plastic wastes as reducing agents or as secondary fuels in some industrial processes. Pyrolysis is a thermal process which enables to obtain a wide range of hydrocarbons with no need of expensive chemical reagents. Plastic wastes are decomposed by the sole effect of temperature in an inert atmosphere, generating liquid and gaseous products which can be useful as fuels and/or sources of chemicals. It has been reported that the use of catalysts in pyrolysis presents some advantages compared to simple thermal processes [1,2]: i.e., lower energy consumption, shorter reaction time and good selectivity to higher added value products.

Over the past two decades, a large number of results on catalytic pyrolysis of plastics have been published in the literature. A wide

range of catalytic materials have been tested: homogeneous catalysts [3,4], acid mesoporous materials [5,6], non-acid mesoporous solids [7,8], FCC catalysts [9,10], zeolites [11,12], metallic oxides [13,14], etc. Zeolites have been found to be particularly efficient catalysts in the degradation of polyethylene [12,15,16]. Most of the previously mentioned studies have been carried out with single plastics or simple mixtures of plastics. Real waste samples or complex mixtures of plastics, which include PVC and PET, have scarcely been studied due to the problems that oxygenated and chlorinated compounds can produce. In this paper, the effect of a zeolite (ZSM-5) in pyrolysis of a complex plastic mixture including PVC and PET (which resembles real plastic wastes) is presented.

On the other hand, from an industrial implementation point of view, the use of expensive catalysts, as those above mentioned, may condition the economy of the process, since high amounts of catalyst would be necessary in a continuous operating plant. Ali et al. [17] concluded that the catalyst cost (type and amount) is a key factor when the economy of both technologies, catalytic cracking and thermal cracking, is compared. Additionally, Cardona and Corma [18] concluded that a plastic waste pyrolysis process could only be supported if the catalyst cost was practically zero. For this reason, the search of “cheap catalysts” for pyrolysis of plastic wastes is of the most interest.

Red Mud, which is a by-product of the alumina production Bayer process, has been used as catalyst mainly to favour hydrogenation reactions, since Fe_2O_3 is its main component [19–22]; however, the presence of other components as SiO_2 , Al_2O_3 or TiO_2 may con-

* Corresponding author. Tel.: +34 946018245; fax: +34 946014179.

E-mail address: alex.lopez@ehu.es (A. López).

tribute to plastic wastes cracking. Up to now, very few studies have been published about the use of Red Mud in pyrolysis [4,23,24].

In a previous work published by the authors [4] a first screening of catalysts in pyrolysis of plastic wastes was performed; the catalysts were tested just at 500 °C and the plastic waste used was a real waste stream which was a very heterogeneous and variable sample, since it came from the rejects of a waste packaging sorting industrial plant. The main conclusion of such study was that it was worthwhile to investigate thoroughly the effect of ZSM-5 and Red Mud on pyrolysis of plastic wastes. For this reason, a thorough study of pyrolysis with these two catalysts is presented in this paper, which includes: (1) a detailed characterization of the catalysts, (2) an exhaustive characterization of the pyrolysis products, and (3) the study of the influence of temperature in the process depending on the catalyst used, with the objective of better understanding the effect of catalysts in the pyrolysis process. Besides, this study has been carried out with a plastic mixture which resembles real wastes but which was prepared by the authors, in order to dispose of a homogeneous and invariable sample which enables to analyse the effect of the catalysts without the distorting effect of the influence of the sample composition.

It is well-known that ZSM-5 zeolite is a suitable catalyst for pyrolysis of plastics while the published information concerning Red Mud is quite limited. The interest of this paper lies mainly in the use of a complex plastic mixture (which resembles real waste streams) and the exhaustive comparison between ZSM-5 zeolite, a very suitable but expensive catalyst, and Red Mud, an inexpensive and scarcely studied material as pyrolysis catalyst.

2. Materials and methods

2.1. Materials

The plastic mixture used for the experiments was composed of 40 wt% polyethylene, 35 wt% polypropylene, 18 wt% polystyrene, 4 wt% poly(ethylene terephthalate) and 3 wt% poly(vinyl chloride). The origin and application of each plastic is showed in Table 1.

The proportions of the plastics in the mixture were established based on the composition of real samples coming from packing and packaging applications which were characterized by the authors in a previous paper [25]. All the materials were used in pellet size (3 mm) for the pyrolysis experiments, while finely ground (<1 mm) samples were prepared for characterization purposes. The characterization results of the plastic mixture are presented in Table 2.

ZSM-5 zeolite, in NH_4^+ cation form and with a $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of 50, was provided by Zeolist International. Red Mud was obtained from a German alumina production company. Its composition is presented in Table 3; it is mainly composed of Fe_2O_3 and Al_2O_3

Table 3

Composition of Red Mud (wt%) (dried basis).

Fe_2O_3	Al_2O_3	TiO_2	SiO_2	CaO	Na_2O	Others ^a
36.5	23.8	13.5	8.5	5.3	1.8	10.6

^a By difference.

together with other oxides including titania. Both catalysts were tested as were received, without any activation operation, and were thoroughly mixed with the plastic sample in a proportion of 10 wt% in all the experiments (10 g of catalyst/100 g of plastics).

3. Experimental

The pyrolysis experiments were carried out in semi-batch conditions in a laboratory scale installation equipped with an unstirred 3.5 dm³ reactor and a condensation-separation system (Fig. 1). In a typical run, 100 g of the sample are placed into the reactor and the system is heated at a rate of 20 °C min⁻¹ up to the desired temperature, followed by an isothermal step of 30 min. The vapours leaving the reactor are swept away by a 1 dm³ min⁻¹ nitrogen flow to a series of gas–liquid separators cooled by running water. The uncondensed products are passed through an activated carbon column and collected as a whole in Tedlar plastic bags, to be afterwards analysed by gas chromatography.

The collected solids (products in the reactor after pyrolysis) and liquids are weighed, and both solid and liquids yields are calculated as weight percentage with respect to the amount of raw material pyrolysed. Gas yields are as a general rule calculated by difference. Some experiments were specifically devoted to directly quantify the amount of gases by gas chromatography; in such experiments a closure of the mass balance of about 90 wt% was obtained. The results of the pyrolysis yields which are presented in Section 4 of this paper are the mean value of at least three pyrolysis runs carried out in the same conditions and which did not differ more than three points in the percentage.

3.1. Analytical techniques

Both the raw materials and the solid and liquid pyrolysis products obtained were thoroughly characterized using the following analytical techniques. The moisture and ash contents of the samples were determined by thermogravimetric analysis according to D3173-85 and D3174-82 ASTM standards respectively, and the elemental composition with an automatic CHN analyser. Chlorine was determined following the method 5050 of the Environmental Protection Agency (EPA) of the United States. The higher heat-

Table 1

Origin and application of the plastics used in the mixture.

Plastic name	Acronymus	Origin	State	Application
Polyethylene	PE	Repsol Química	Virgin	Household
Polypropylene	PP	Repsol Química	Virgin	General
Polystyrene	PS	Dow Chemical	Virgin	General
Poly(ethylene terephthalate)	PET	Remaplast S.A. ^a	Recycled	Bottles
Poly(vinyl chloride)	PVC	Gaiker ^b	Recycled	Bottles

^a Spanish Company devoted to municipal plastics recycling.

^b Spanish Technology Centre dedicated to research and innovation.

Table 2

Moisture, ash, elemental composition (wt%) and HHV (MJ kg⁻¹) of the plastic mixture pyrolysed.

Moisture	Ash	C	H	N	Cl	Others ^a	H/C ratio	HHV
0.1	0.0	84.7	12.5	<0.1	1.1	1.5	1.8	43.9

^a By difference.

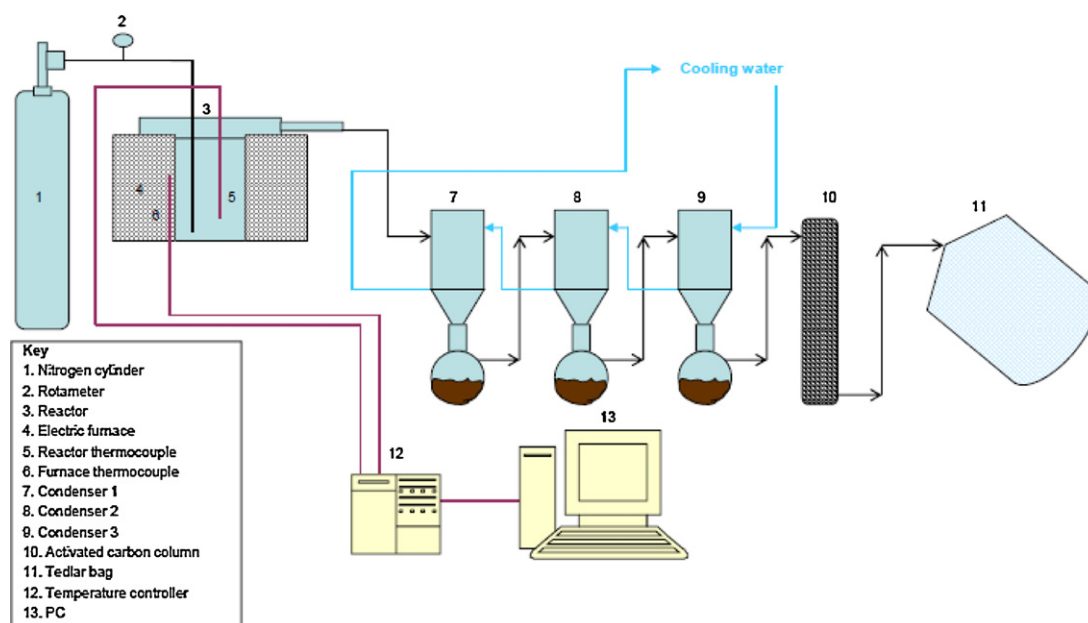


Fig. 1. Flow sheet of the experimental set-up used.

ing value (HHV) was determined with an automatic calorimetric bomb.

Additionally, pyrolysis liquids were also analysed by gas chromatography coupled with mass spectrometry detector (GC–MS), Agilent 6890 and Agilent 5973 respectively. Characteristics of the method used are shown in Table 4. Identification of the constituents was based on comparison of the retention times with those of calibration samples and on computer matching against commercial library of mass spectra (Wiley7n) and MS literature data. The library-matched species which exhibited a degree of match lower than 85% were classified as “Not identified”.

Pyrolysis gases were analysed by means of a gas chromatograph coupled with thermal conductivity and flame ionization detectors (GC–TCD/FID). Table 5 shows the characteristics of the method used. Due to the difficulty in distinguishing among isomers from C3 to C6, such discrimination has not been made. The HHV of the

gases was calculated according to their composition and to the HHV of the individual components [26].

The textural properties of the catalysts were determined by means of nitrogen adsorption–desorption isotherms at 77 K in AUTOSORB-1 Quantachrome equipment. Surface areas were calculated by means of BET equation and external surface areas and micropores volume were obtained applying the *t*-plot method. Total pore volume was measured at $P/P_0 = 0.99$.

Acidity of the catalysts was measured by temperature-programmed desorption (TPD) of ammonia on a Micromeritics AutoChem 2910 instrument. Prior to adsorption experiments, the samples (86–300 mg) were first pre-treated in a quartz U-tube in nitrogen stream at 500 °C. Then, they were cooled down at 100 °C in N₂ flow (20 cm³ min^{−1}) before the ammonia adsorption started. The adsorption step was performed by admitting small pulses of ammonia in Ar at 100 °C up to saturation. Subsequently, the samples were exposed to a flow of argon (50 cm³ min^{−1}) for 2 h at 100 °C in order to remove reversibly and physically bound ammonia from the surface. Finally, the desorption was carried out from 100 to 500 °C at a heating rate of 10 °C min^{−1} in Ar stream (50 cm³ min^{−1}). This temperature was maintained for 15 min until the adsorbate was completely desorbed. The amount of ammonia desorbed at a given temperature range was taken as a measure of the acid site concentration, whereas the temperature range at which most of the ammonia was desorbed indicated the acid strength distribution [27,28]. SEM images were obtained in a LEO 1525 electron microscope.

4. Results and discussion

4.1. Catalyst characterization

Fig. 2 shows the N₂ adsorption–desorption isotherms of the catalysts. According to the International Union of Pure and Applied Chemistry (IUPAC) classification of adsorption isotherms, the shape of the adsorption isotherm for ZSM-5 is a combination of the type I (Langmuir) isotherm and type IV isotherm. The adsorption volume at a very low relative pressure ($P/P_0 < 0.1$) is high, indicating the presence of microporous adsorption (type I). With the increase in relative pressure, capillary condensation occurs, showing type IV

Table 4
GC–MS method characteristics.

Column	HP5MS
Carrier gas	He
Carrier gas flow	1.0 mL/min
Initial temperature/initial time	40 °C/10 min
Heating rate	8 °C/min
Final temperature/final time	280 °C/10 min
Injection temperature	280 °C
Injection volume	1.0 µL (split 100:1)
Detector temperature (quad/source)	150 °C/230 °C

Table 5
GC–TCD/FID method characteristics.

Columns	Molecular sieve 13X Chromosorb 102
Carrier gas	He/Ar
Carrier gas flow	48 mL/min
Initial temperature/initial time	40 °C/10 min
Heating rate	6 °C/min
Final temperature/final time	200 °C/10 min
Injection temperature	110 °C
TCD detector temperature	110 °C
FID detector temperature	200 °C

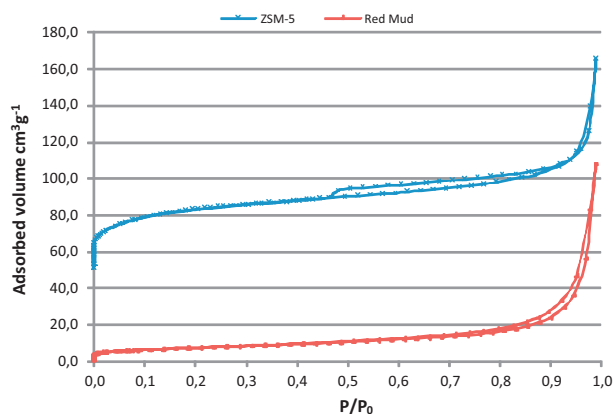


Fig. 2. N₂ adsorption-desorption isotherms at 77 K of ZSM-5 zeolite and Red Mud.

Table 6

Textural properties of the catalysts used.

Property	ZSM-5 zeolite	Red Mud
BET surface area (m ² g ⁻¹)	412.0	27.49
External surface area (ESA) (m ² g ⁻¹)	65.88	25.78
Micropore area (m ² g ⁻¹)	346.1	1.71
Micropore volume (MPV) (cm ³ g ⁻¹)	0.100	0.009
Total pore volume (TPV) (cm ³ g ⁻¹)	0.397	0.184

behaviour. The hysteresis loop, which resembles the H4 type, may be attributed to the mesoporous structure. This type of hysteresis is usually found in solids consisting of aggregates or agglomerates of particles forming slit shaped pores (plates or edged particles like cubes), with uniform size and/or shape. Additionally, the steep jump observed at high relative pressures (>0.9) indicates some macropores existence.

Concerning Red Mud, the adsorption isotherm could suggest that this catalyst is a meso-macroporous material, since the adsorption is very low at low pressures and starts fast increasing from $P/P_0 \approx 0.7$.

Table 6 shows the textural properties of the catalysts. Both catalysts present significant differences in pores distribution. BET surface area of ZSM-5 zeolite is 412 m² g⁻¹, with high micropore area (346.1 m² g⁻¹) and also high micropore volume (0.100 cm³ g⁻¹), which indicates the existence of relatively high internal porosity in this catalyst. On the other hand, Red Mud shows BET and ESA of 27.49 and 25.78 m² g⁻¹ respectively, which means that in addition to being a low porosity material, most of the pores are macro and mesopores located in the external surface of the catalyst. The BET surface area obtained with Red Mud agrees with the expected value for this material, which has been reported to be in the range 20–30 m² g⁻¹ [29,30].

Fig. 3 shows the NH₃ TPD plot of both catalysts. ZSM-5 catalyst shows the typical TPD profile of this zeolite, which consists in two main desorption peaks which prove the presence of both weak and strong acid sites [31,32], the latter corresponding to Brönsted type acid sites [33]. The total acidity of the catalyst was determined to be 0.176 mmol NH₃ g⁻¹ and the quantified area under the two peaks showed a distribution of 37% weak acidity and 63% strong acidity. On the other hand, as expected according to the composition of Red Mud, lower total acidity was obtained for this catalyst (0.094 mmol NH₃ g⁻¹) with 26% weak acidity and 74% strong acidity, which was quantified under an intense peak with a very slow NH₃ desorption at 550 °C. Wang et al. [30] suggested that the desilication product in Red Mud structure, containing Si–O–T units (T = Si or Al), confers this catalyst a zeolitic nature, which could be responsible for the strong acidity measured in this paper.

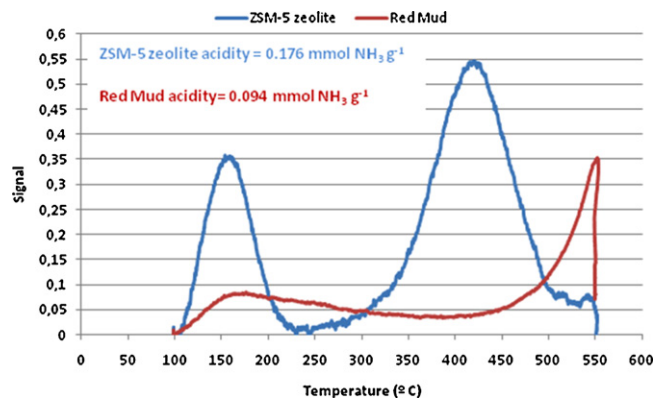


Fig. 3. NH₃ TPD plot of ZSM-5 zeolite and Red Mud.

SEM images of both catalysts are presented in Fig. 4. ZSM-5 zeolite is formed by uniform crystals with sizes within the nanometer size (≈ 100 – 300 nm), while Red Mud is constituted by different size particles and crystals with also different shapes and morphology, ranging from nanometer size particles to quite big ones formed by agglomeration of the former, which confers it a highly heterogeneous internal structure.

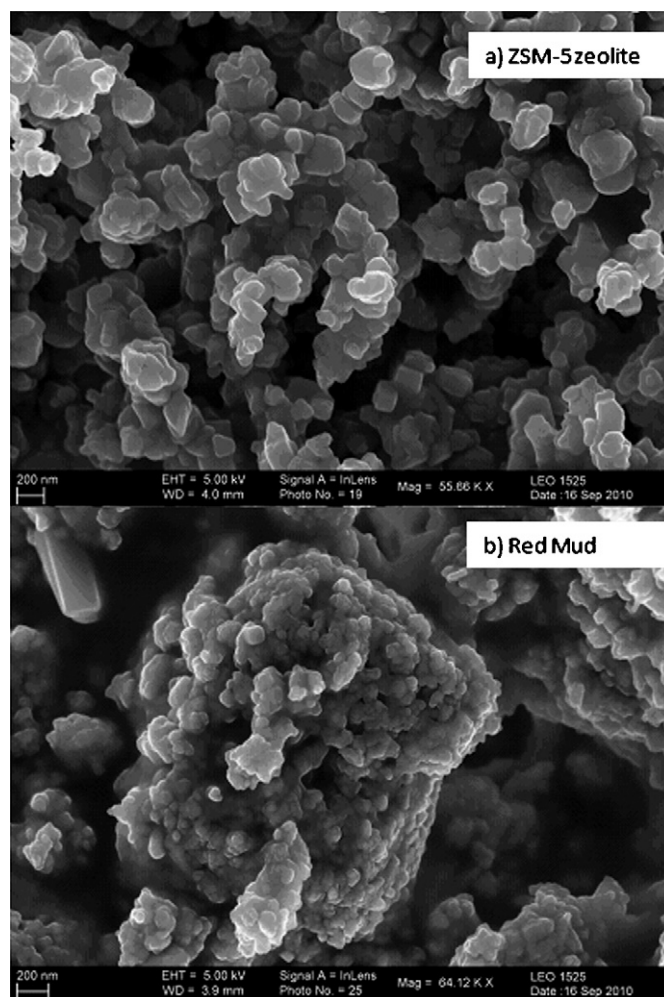


Fig. 4. SEM micrographs of ZSM-5 zeolite (a) and Red Mud (b).

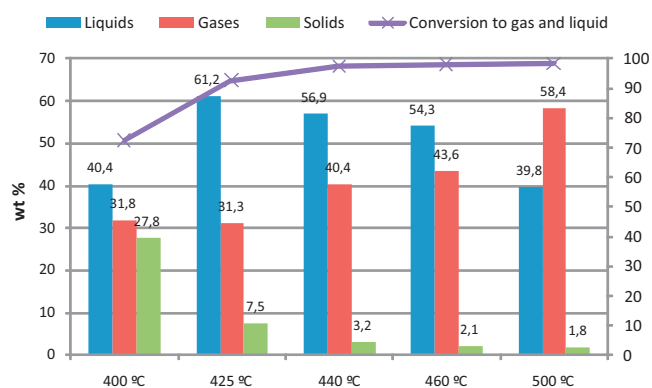


Fig. 5. Pyrolysis yields (wt%) as a function of temperature in the ZSM-5 catalytic experiments.

4.2. Influence of temperature on ZSM-5 catalytic pyrolysis yields

A preliminary study of the effect of temperature on catalytic pyrolysis yields was carried out in order to select the most appropriate temperatures to compare the behaviour of both catalysts. The range between 400 and 500 °C was explored using ZSM-5 catalyst. The pyrolysis yields obtained (weight %) are presented in Fig. 5, as well as the conversion to liquid and gas. As it can be seen, 400 °C is a temperature not high enough to decompose the entire sample, since a solid residue (unconverted plastic) near 30 wt% is obtained. On the contrary, at 425 °C more than 90 wt% conversion is achieved, and from 440 to 500 °C the solid fraction yield can be considered more or less constant. At this temperature range, the remaining solid fraction was mainly composed of the char that is produced in pyrolysis of many polymeric materials [e.g. 34–36]. Therefore, 440 °C is a temperature high enough to achieve total conversion of the plastic mixture to gaseous and liquid products when ZSM-5 zeolite is used.

It can also be seen that as temperature is raised from 425 to 500 °C, the liquid yield decreases and consequently the gas production increases. This behaviour has also been reported by many other authors [1,31,37] and it is attributed to the stronger cracking of C–C bonds that takes place at higher temperatures, which gives rise to lighter hydrocarbons with shorter carbon chains. Among the temperatures at which total conversion is achieved, 440 °C is the lowest one and at the same time the one that yields the greatest amount of liquids, while 500 °C is the temperature that leads to the highest gas yield. For this reason, these temperatures (440 and 500 °C) were selected as the most interesting to compare thermal and catalytic (ZSM-5 and Red Mud) pyrolysis.

4.3. Comparison among thermal, ZSM-5 and Red Mud pyrolysis

4.3.1. Pyrolysis yields

The yields of liquid, gaseous and solid products (weight %) obtained at 440 and 500 °C in the thermal and catalytic runs are presented in Fig. 6. As it can be seen, the liquids are the main frac-

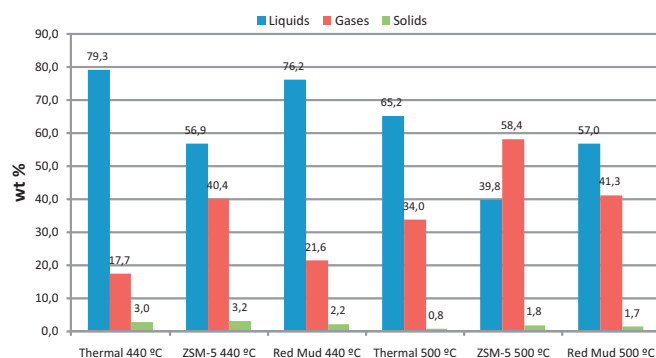


Fig. 6. Pyrolysis yields (wt%) as a function of temperature and catalyst used.

tion in thermal pyrolysis, reaching 79.3 wt% at 440 °C and 65.2 wt% at 500 °C. The gas fraction is very low at 440 °C (17.7 wt%) but higher at 500 °C (34.0 wt%), due to the previously mentioned stronger cracking that is produced at higher temperatures. When ZSM-5 zeolite is added to the sample, a great gas production enhancement is observed at 440 °C (40.4 wt% gases and 56.9 wt% liquids) and even higher at 500 °C (39.8 and 58.4 wt% respectively). This significant increase in gas yield produced by ZSM-5 zeolite, which has also been reported by other authors [16,31,38], demonstrates the cracking ability of this catalyst, which is a consequence of its high porosity and strong acidity, as shown in Fig. 3.

The influence of Red Mud in pyrolysis yields is lesser than that of the zeolite; slightly higher gas yields than without catalysts are obtained at both 440 and 500 °C (4 and 7 wt% higher respectively), which indicates that its cracking ability is lower than that of the zeolite, as was expected according to its lower surface area and acidity. Sushil and Batra [29] reported that in hydrogenation and liquefaction applications Red Mud becomes active at 400 °C, which may be the reason why higher increase in gases is obtained at 500 °C than at 440 °C.

The difference in the pyrolysis yields obtained with Red Mud and ZSM-5 zeolite could be explained as follows. Taking into account that the type of contact among catalysts and plastics is solid–solid and comparing the product yields of Fig. 6, it could be suggested that thermal pyrolysis is the first step in the reaction since pyrolysis liquids yield is higher without catalysts at both temperatures. The thermally degraded products could then more easily access to the acid sites into the catalysts porous structure, leading to higher cracking rate (more gases and fewer liquids than in the thermal runs). If the process takes place according to this theory, the additional conversion produced by the catalysts is directly related to their textural and acid properties. Consequently, the higher porosity and acid strength of ZSM-5 zeolite compared to those of Red Mud, produce more gases and fewer liquids than the latter.

Fig. 6 also shows that the solid yields obtained at 440 °C are in every case a little bit higher than those obtained at 500 °C, which implies that at 440 °C some unconverted product remains in the solid fraction mixed with char. The solid yield of the thermal run at

Table 7

Main components of the pyrolysis liquids determined by GC–MS (% area).

Compound	Thermal 440 °C	ZSM-5 440 °C	Red Mud 440 °C	Thermal 500 °C	ZSM-5 500 °C	Red Mud 500 °C
Toluene	8.1	12.3	8.9	8.1	17.5	14.2
Dimethyl-heptene	7.8	1.8	7.8	5.9	0.8	2.1
Ethyl-benzene	5.7	10.6	6.6	5.0	9.6	9.0
Xylenes	n.d. ¹	10.1	1.4	<3.0	13.8	4.2
Styrene	46.3	31.4	46.0	48.4	27.9	42.3
α-Methyl-styrene	3.2	4.5	4.3	4.2	3.4	4.9
Methyl-naphthalene	n.d. ^a	3.3	0.7	n.d. ¹	3.5	2.3

^a Not detected.

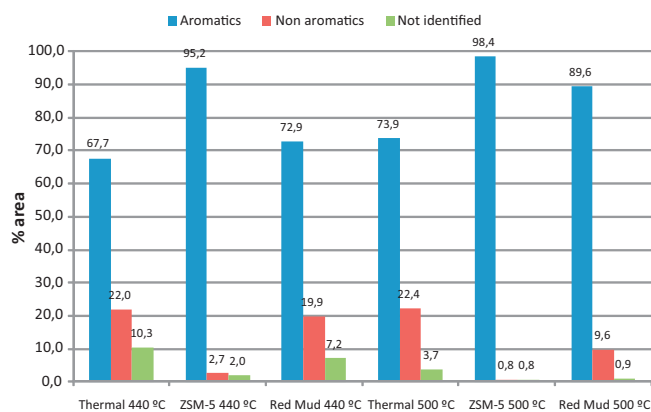


Fig. 7. Aromatics, non-aromatics and not identified compounds in the liquid fractions (% area).

500 °C is lower than those of the catalytic runs, which may be due to the fact that char formation is enhanced by acid catalysts, as it has been reported by other authors [5,39].

4.3.2. Pyrolysis liquids

The characterization results of the pyrolysis liquids are presented in Figs. 7 and 8 and Tables 7 and 8. In Fig. 7 total aromatics, non-aromatics and unidentified compounds have been quantified; Fig. 8 shows the pyrolysis liquids compounds grouped according to their carbon number, while the percentage of their main components have been summarized in Table 7. Additionally, the elemental composition and higher heating value (HHV) of the pyrolysis liquids obtained in each experiment are presented in Table 8. Fig. 7 shows that aromatic compounds are the main fraction of the liquids derived from all the experiments at both temperatures. However, it must be mentioned that the values presented in Tables 7 and 8 and

Table 8

Elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis liquids.

Experiment	C	H	Cl	Others ^a	H/C ratio	HHV
Thermal 440 °C	86.1	12.3	0.2	1.4	1.7	44.7
ZSM-5 440 °C	87.6	11.2	1.2	0.0	1.5	42.2
Red Mud 440 °C	85.4	11.8	0.5	2.2	1.7	44.3
Thermal 500 °C	86.5	11.3	0.5	1.5	1.6	43.3
ZSM-5 500 °C	88.8	9.9	1.3	0.0	1.3	40.6
Red Mud 500 °C	87.5	10.5	1.1	0.9	1.4	40.9

^a By difference.

Figs. 7 and 8 are % area and not wt%, thus the relative amounts of aliphatic and aromatic products are not straight reflected by these values since different kinds of organic compounds have different responses in MS.

At first sight, a general increase in aromatic compounds is observed when temperature is increased from 440 to 500 °C. Therefore, temperature is also a key parameter when catalyst is used, producing more aromatics as a consequence of secondary reactions which are produced to a major extent at high temperatures [40]. The use of ZSM-5 zeolite produces a strong aromatization of the liquid products, which reach 95.2 and 98.4% area aromatics at 440 and 500 °C respectively, while just 67.7 and 73.9% area aromatics are obtained in the same conditions without catalyst. These results show that ZSM-5 zeolite has a great influence in aromatization reactions. Marcilla et al. [41] reported that the high percentage of aromatics obtained with this zeolite could be attributed to the high number of Brönsted acid sites contained in its structure, where aromatization reactions are favoured. Therefore, it can be stated that the strong aromatization obtained with ZSM-5 zeolite in this study is related to the strong acidity of the zeolite used (63% of strong acid sites). It is known that strong acid sites contribute to coke-forming reactions, leading to the fast deactivation of the catalysts. In this sense, the authors have carried out a parallel study concerning the deactivation and regeneration of the ZSM-5 zeolite. Such study has proved that the catalyst is strongly deactivated after one pyrolysis run, but it can be easily regenerated by means of coke combustion.

The aromatics content obtained with Red Mud at 440 °C is only slightly higher than that obtained in the thermal process, but at 500 °C the aromatics content is considerably higher in presence of Red Mud compared to the thermal run. This is in agreement with the theory that Red Mud needs high temperatures to be activated. In fact, the aromatic production reached at 500 °C was 89.6% area, which is 15% higher than that obtained in the thermal run. These results are not in accordance with those reported by Yanik et al. [23] in a work about the catalytic effect of Red Mud in the degradation of a polymer mixture, since they concluded that Red Mud did not show any effect in the cracking process; on the contrary, similar results concerning aromatics formation were obtained by the authors in a previous paper in which plastics were pyrolysed in presence of Red Mud [4]. These differences may be attributed to the fact that the composition of Red Mud varies depending of the specific process from which it is obtained. It has been reported [42] that some Red Mud components such as Na₂O may act as poison, diminishing the catalytic activity. Different Na₂O contents may be the reason for the differences in Red Mud activities found by different authors. In fact, the Red Mud used by Yanik et al. [23] contained a higher proportion of Na₂O (7.1 wt%) than that of the Red Mud of the present study (1.8 wt%).

Fig. 8 shows the pyrolysis liquids compounds grouped according to their carbon number. As it can be seen in Fig. 8a, the pyrolysis liquids obtained at 440 °C are a mixture of C7–C20 compounds in the thermal experiment, of C7–C16 compounds in the ZSM-5 experiment and of C7–C19 compounds in the case of Red Mud. Therefore the cracking ability of ZMS-5 zeolite is also shown in



Fig. 8. Pyrolysis liquids characterized by carbon number (% area) at 440 °C (a) and 500 °C (b).

Table 9
GC–TCD/FID analysis (wt%) and HHV (MJ kg^{−1}) of pyrolysis gases.

Experiment	Thermal 440 °C	ZSM-5 440 °C	Red Mud 440 °C	Thermal 500 °C	ZSM-5 500 °C	Red Mud 500 °C
H ₂	0.1	0.8	0.7	0.4	1.0	0.8
CO	0.4	0.7	0.5	0.7	0.7	1.2
CO ₂	2.7	4.9	8.5	2.9	1.9	7.5
Methane	9.7	8.8	11.2	8.3	10.6	14.1
Ethane	12.7	6.5	6.3	10.0	8.6	12.4
Ethene	8.8	9.3	8.1	12.2	11.1	17.3
C3–C4	43.8	57.0	39.2	46.7	50.7	41.5
C5–C6	21.7	11.9	25.5	18.7	15.5	5.3
HHV	48.7	46.0	49.1	48.6	49.8	47.1

pyrolysis liquids, leading to lighter compounds production than in thermal decomposition. Similar results have been obtained by other authors before [43,44]. In all cases, most of pyrolysis liquids compounds are in the C7–C10 range, with a sharp peak in C8 fraction; this is a rather convenient result from the point of view of potential applications of these liquids, since C5–C10 is the gasoline carbon number range. A small peak can be also detected in C15–C16 fractions, which corresponds to substituted double ring aromatic compounds. Similar profiles can be found in Fig. 8b, which shows the distribution of pyrolysis liquids compounds at 500 °C. Again, higher Red Mud activity can be observed at this temperature, shortening the pyrolysis liquids carbon range at same extent as the zeolite (C7–C16), while thermal pyrolysis liquids reach C19 compounds.

The main individual components of the pyrolysis liquids are presented in Table 7. For the sake of reduction, only those compounds with a percentage quantified area greater than 3% have been included. Styrene is the most abundant product with percentage areas ranging from 27% to almost 50% area. The next abundant products are toluene (8–17.5% area) and ethyl-benzene (5–10.6% area), and other C8 aromatic compounds like xylenes and α -methylstyrene, which justifies the sharp peak observed in C8 in Fig. 8. It is somewhat surprising that so much styrene is obtained in the pyrolysis liquids, since the initial sample is mainly composed of polyolefins (40 wt% PE and 35 wt% PP) and only contains 18 wt% PS. The authors have confirmed that such GC–MS peak corresponds to styrene using the following procedure: (1) comparison of the mass spectrum of this peak with the theoretical mass spectrums of styrene found in two different databases (Wiley7 and NIST), (2) analysis of pure styrene (standard quality) with the same conditions in order to compare the mass spectra and (3) analysis of the liquids with multidimensional chromatography (GCxGC). All the mass spectra (i.e. the databases ones, that of pure styrene and that of the peak assigned to styrene in the GC–MS analysis of the liquids, both in mono and multidimensional GC–MS), were almost coincident.

In general terms, the production of styrene is higher in the thermal experiments than when catalyst is used. This effect is more pronounced with ZSM-5 zeolite than with Red Mud. At the same time, ZSM-5 zeolite enhances the production of toluene, ethyl-benzene and xylenes. Serrano et al. [5] explained that in the catalytic decomposition of PS, the carbenium nature of the acid catalyzed cracking leads to aromatic products different from styrene, while styrene is the main product in thermal decomposition of PS. The high proportions of styrene obtained in the thermal experiments of this work suggest that styrene is probably formed by other mechanisms (secondary reactions) apart from direct PS decomposition, but the decrease of this compound in the presence of ZSM-5 zeolite could be explained by the above mentioned theory reported by Serrano et al. [5].

Concerning Red Mud, no significant differences with the thermal experiment were found at 440 °C, but there did were differences in the experiments carried out at 500 °C: lower styrene production

and higher ethyl-benzene and toluene yields were observed. It is not easy to propose a mechanism to explain these changes, due to the fact that Red Mud is a complex mixture of metal oxides which can influence in different ways in the process. On the one hand, Al₂O₃ and SiO₂ could enhance cracking due to their acid nature, leading to a decrease in styrene yield as it has been exposed above for ZSM-5. However, on the other hand, Fe₂O₃, which is known to have hydrogenation activity, is the main component of Red Mud. The hydrogenation character of Red Mud could be even stronger than that of Fe₂O₃ alone due to the presence of TiO₂ impurity, as it has been reported by Sushil and Batra [29]. If such is the case, the increase in ethyl-benzene could have been produced by direct hydrogenation of styrene, while the increase in toluene could have been produced by hydrocracking of styrene, yielding toluene and methane.

The elemental composition and higher heating value (HHV) of the pyrolysis liquids are presented in Table 8. It can be seen that these liquids are mainly composed of carbon and hydrogen, with small quantities of chlorine which comes from PVC. The lowest H/C ratios are obtained with ZSM-5 zeolite (1.5 at 440 °C and 1.3 at 500 °C) which is in agreement with the greater aromatics content of the liquids produced by this catalyst, as it has been observed in the GC–MS analyses. It is worth mentioning that pyrolysis oils have very high HHV, similar to those of conventional liquid fuels, so they may be considered as an appropriate alternative to fossil fuels. However, it has to be mentioned that the chlorine contained in the liquids could be very detrimental for their application. For this reason, the authors have studied, in a previous work [45], different alternatives to reduce the chlorine content of the pyrolysis liquids and found out that it can be reduced to a high extent if a low temperature dehydrochlorination step is applied prior to the pyrolysis process.

4.3.3. Gas composition

Table 9 shows that pyrolysis gases are composed of hydrocarbons ranging from C1 to C6, hydrogen and some carbon dioxide and monoxide. It can be seen that C3–C4 is the main fraction of pyrolysis gases both in thermal and in catalytic pyrolysis. ZSM-5 zeolite increases this fraction especially at 440 °C (from 43.8 wt% without catalyst to 57.0 wt%). This effect of ZSM-5 zeolite has also been reported by other authors [46,47] and it could be related to the shape selectivity of this catalyst (pore size). Once more the effect of Red Mud is noticeable at 500 °C, while it is negligible at 440 °C. At 500 °C Red Mud produces larger quantities of light hydrocarbons (C1–C2) and diminishes the proportion of C3–C6 gases. It is worth noting that Red Mud generates the highest proportion of methane at 500 °C (14.1 wt%), which may be caused by the hydrocracking reaction of styrene (to lead toluene and methane), which has been previously mentioned, proposed in the GC–MS liquid composition discussion (Table 7). Additionally, the great proportions of CO₂ obtained with Red Mud may be attributed to the catalytic activity of Fe₂O₃, since it has been reported [42] that the formation of CO₂ is produced by the stepwise reduction of haematite (Fe₂O₃)

Table 10Moisture, elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis solids.

Experiment	Moisture	C	H	Cl	Others ^a	H/C Ratio	HHV
Thermal 440 °C	n.d. ^b	86.6	10.0	<0.1	3.3	1.4	42.7
ZSM-5 440 °C	0.7	23.0	2.2	0.1	74.0	1.1	13.1
Red Mud 440 °C	0.8	13.6	0.7	4.2	80.7	0.6	11.3
Thermal 500 °C	0.2	93.7	3.5	0.3	2.3	0.4	38.2
ZSM-5 500 °C	1.2	13.8	0.9	0.1	84.0	0.8	5.9
Red Mud 500 °C	0.9	13.9	0.7	3.7	80.8	0.6	6.7

^a By difference.^b Not determined.

to wustite (FeO) and metallic iron (Fe) by means of the reaction between hydrocarbons and Fe₂O₃.

It has to be mentioned that both catalyst enhance hydrogen production, which could be attributed to the hydrogen abstraction that takes place during aromatization reactions. Finally, it is also worth noting that the HHV of the gases in all cases is in the range of that of natural gas (48–53 MJ kg⁻¹). For this reason pyrolysis gases may be used as gaseous fuels to supply the energetic demand of the process, and the surplus may be valorised.

4.3.4. Pyrolysis solids

The composition of the pyrolysis solids is presented in Table 10. The great differences among the thermal and the catalytic runs derived solids are due to the fact that the catalyst is mixed with the pyrolysis solids. The higher carbon and lower hydrogen contents of the solid obtained in the thermal run at 500 °C (93.7 and 3.5 wt%) in comparison with that obtained at 440 °C (86.6 and 10 wt%) is due to the fact that some unconverted product remained in the solid fraction in the 440 °C experiment. Somehow similar behaviour can be observed in the ZSM-5 experiments, while in the Red Mud runs the composition of the solids obtained at 440 and 500 °C are almost coincident. These results are in agreement with the data in Fig. 6, where it can be appreciated that for thermal and ZSM-5 experiments the solid yields are higher at 440 °C than at 500 °C, showing further degradation, while the solid yield of Red Mud experiments are similar at 440 and 500 °C. Anyway, the pyrolysis solids are in all cases carbonaceous materials with high HHV. They could find applications such as alternative to fossil solid fuels, pigment or carbon black after catalysts separation and recovery.

5. Conclusion

Pyrolysis is a promising alternative for plastic wastes recycling. Poor quality mixed plastics can be converted into valuable liquids with high HHV and significant proportions of chemicals as styrene, ethyl-benzene or toluene; they can be used as an alternative to fossil fuels or as a source of chemicals for petrochemical processes. Additionally, C₃–C₄ rich gases with very high HHV are generated, which can be used for power generation for the process and/or for external applications. Pyrolysis solids are a minor by-product of the process mainly composed of carbon, with several potential applications (fuel, pigment, etc.).

The use of catalysts in pyrolysis of plastic wastes has a significant influence in both products yields and characteristics of the products. ZSM-5 zeolite is a very active catalyst for plastic waste pyrolysis; on the one hand, it promotes gas production and give rise to lower molecular weight and higher aromatic containing liquids even at low temperatures (440 °C), despite the fact that the initial sample is mostly polyolefinic. On the other hand, it enables to operate at lower temperatures (440 °C), achieving similar yields and product properties than in thermal runs at higher temperatures (500 °C). This implies energy saving and consequently operating costs reduction.

Red Mud shows a noticeable activity in plastic waste pyrolysis. However, it needs higher temperatures than ZSM-5 zeolite to become active in the pyrolysis process. It shows no activity at 440 °C while at 500 °C it increases gas yields, decreases liquids viscosity and promotes liquids aromatization. Taking into account these effects and considering that Al₂O₃ and SiO₂ are components of Red Mud and that it has both weak and strong acid sites in its structure, Red Mud may be considered a solid acid catalyst. However, there are differences in the nature of the liquids and gases obtained with Red Mud and ZSM-5 zeolite, which imply that other reaction pathways, such as hydrogenation and/or hydrocracking reactions take place, probably due to the presence of Fe₂O₃ and TiO₂ among the Red Mud components. The fact that Red Mud, which is an inexpensive by-product of the alumina industry, could help recycling of municipal plastic wastes by pyrolysis is of the most interest not only from the point of view of the economy of the process, but also from a global recycling point of view, since Red Mud is in its turn an industrial waste.

Acknowledgements

The authors thank the Spanish Ministry of Education and Science (MEC) (CTQ 2007-67070/PPQ) as well as the Basque Country Government (ETORTEK 2007 IE07-207, GIC 07-09-IT-354-07 and Researchers' Formation Program – 2008) for financial assistance for this work.

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